From headlines to portfolio risk

A unified AI/NLP risk engine with Module B: strategic event-driven stress testing.

CANDIDATE

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PROBLEM

Risk analysts must connect unstructured financial news to portfolio exposures quickly and traceably.

APPROACH

Real publisher news -> structured risk signals -> hypothetical shocks -> before/after portfolio estimates.

DEMO SCOPE

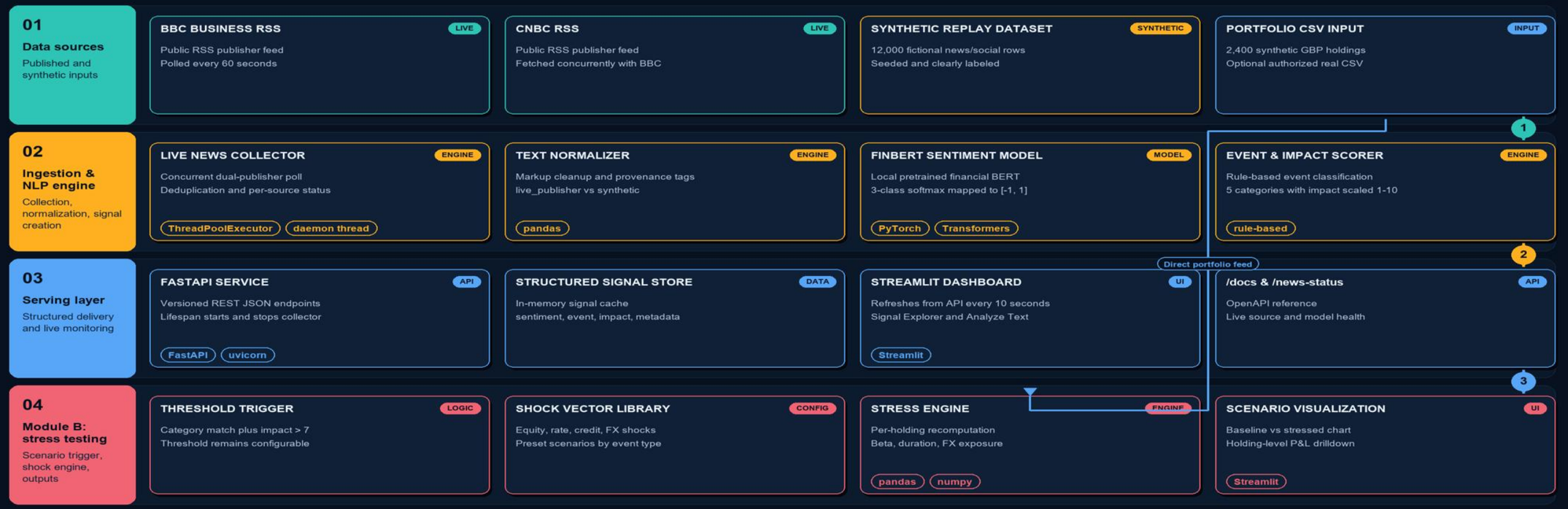
A locally runnable Python prototype. No investment advice, production valuation or guaranteed predictions.

One engine, modular applications

FastAPI | Streamlit | PyTorch + Transformers | pandas | RSS background worker

SIGNAL & STRESS | SYSTEM ARCHITECTURE

End-to-end data flow: real and labeled-synthetic ingestion, local FinBERT scoring, structured API delivery, and event-driven portfolio stress testing



FLOW

1. Ingestion normalizes real and synthetic text and preserves provenance.
2. FinBERT and rules convert text into sentiment, event, and impact signals.
3. Signals are exposed via JSON and can trigger stress testing when severity is high.

LEGEND

- Real published data
- Synthetic or rule-based processing
- Computed or served signal
- Stress-testing application output

RSS polling is not publisher push streaming. Synthetic data remains labeled at schema level. Scenario losses are illustrative, not forecasts.

Dataset used: real + clearly fictional

Self-generated synthetic data is permitted by the supplied dataset clarification.

REAL INPUTS

BBC and CNBC RSS: up to 25 headlines each, collected every 60 seconds. URLs and publication times retained.

DETAILED SYNTHETIC DATA

12,000 fictional news/social records; 2,400 holdings; 240 issuers; 5 asset types, 6 sectors and 4 regions.

REPRODUCIBLE DEMO

Seed 20261008; CSV/JSONL datasets and generation manifest. Optional replay emits 6 events every 10 seconds.

ASSUMPTIONS

GBP values and bounded sensitivities are fictional. No actual bank transactions, calibrated distributions or live social posts.

Explainable workflow, honest model scope

Pretrained sentiment is active; event and severity rules remain illustrative.

SIGNAL CONTRACT

Sentiment [-1, 1], event category, impact [1, 10], source, timestamp and provenance available through JSON APIs.

NLP VERIFICATION

Official local FinBERT: positive +0.84, negative -0.82, neutral 0.00 on three smoke-test sentences, not a benchmark.

FAST INSPECTION

Six editable cases, source/category filters, downloads, cue highlights and explicit stress-threshold explanations.

BOUNDARIES

Keyword cues are not FinBERT attributions. Impact is not a calibrated prediction; negation and factual accuracy are not resolved.

Module B: quantify hypothetical stress

A geopolitical scenario illustrates the baseline-to-stressed portfolio workflow.

EVENT-DRIVEN TRIGGER

Risk category and impact drive the scenario. Default threshold >=7; select 8 for the example impact >7.

ILLUSTRATIVE SHOCKS

Geopolitical: equity -10%, rates +2 percentage points, spreads +1.5 points, FX -5%.

BUNDLED DATASET RESULT

GBP 14,685,727,454.43 baseline -> GBP 13,392,991,408.98 stressed value: approximately -8.80%.

INTERPRETATION

Holding P&L and asset-class charts support risk review. Linear sensitivities and zero-clipping are simplified; scenarios are independent.

Validated prototype, reviewer-ready demo

Business value: traceability from a published headline to an estimated portfolio exposure change.

VERIFICATION

37 tests passed at last validation. Both real publishers, trained inference and unattended collection were verified.

REVIEWER WALKTHROUGH

Confirm model/source status -> load Debt default -> analyze cues -> inspect stress impact -> open Dataset studio and API docs.

RUN LOCALLY

Install requirements; generate datasets; prepare official FinBERT; run python run_demo.py. Dashboard: localhost:8501.

NEXT STEPS & SUBMISSION

Evaluate unseen headlines and calibrate impact. Public repo: vit-sharan-r-hackathon (pending). Demo video URL: pending.